No. 9-Dars

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Model improvement → Bu machine learning modelining bashorat qilish natijalarini yaxshilash ya'ni modelning prediction qiymatlarini haqiqiy qiymatlarga yaqinlashtirish.

Regressionda MSE MAE larni kamaytirib R_2 scoreni oshirish model improvement.

Classificatsionda accuracy scoreni oshirish.

Yaxshi model → Yaxshi dataset + aqilli va foydali features + to'g'ri algorithmlar + tuning + overfitting - ni oldini olish.

Transform features → Input featurelarni ML algo-ritmlar uchun moslashtirish jarayoni yani featurelarni kompyuterga osonroq tilda tushuntirish ya'ni featurelarni o'zgartirish.

Transform features Turlari

- Feature Scaling
- Log transformation
- Power transformation
- Polynominal Features
- Encoding

Parametr nima? → Model training yordamida o'rganadigan qiymatlar.

Hyperparameter → Modelni o'rgatishdan oldin biz belgilaydigan sozlamalar.

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~~Ensemble Methods~~

Ensemble Methods → Bir nechta model yoki algoritm natijalarini birlashtirib, umumiy bashorat olish bitta modelga ishonib qolmay bir nechtasidan foydalanish.

Bagging → Bir nechta model turli na'munalar asosida parallel o'rgatiladi va natijalari birlash-tiriladi. (Random Forest)

Boosting → Modellar ketma ket o'rgatiladi. Keyingi model oldingi modellar xatolarini kamaytirishga harakat qiladi.

Voting → Bir nechta classification modelining bashoratlari ovoz berish orqali birlashtiriladi.

Log Transformation → feature qiymatlariga logarifmik matematik amal qo'llash orqali ularning taqsimotini o'zgartirish jarayoni.

Skewness nima → datasetdagi qiymatlarning taqsimoti qanchalik simmetrik yoki qiyshaygan ekanligini ko'rsatuvchi statistik ko'rsatkich.

Future Engineering techniques.

1 Missing value

2. Encoding

3. Scaling

4. Feature creation

5. Feature selection

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Model Improvement yollari.

1. Data quality and quantity.
2. Feature Engineering
3. Scaling
4. Turli xil algoritmlar bilan tekshirish.
5. Hyperparametr tuning
6. Cross validation.
7. Regularization
8. Ensemble methods.
9. Handling Imbanced Dataset.

Log Transformation.

```
df = preprocessed_data
num_col = df.select_dtypes(include=['float64'])
num_col
num_col.skew()
import numpy as np
skewness = df.select_dtypes(include=['float64']).skew()
log_transformation = skewness[skewness >= 0.5)].index.tolist()
log_transformation
for col in log_transformation:
df[col + 'log'] = np.log1p(df[col])
df.info()
df.columns.
```